A proposal of requirements validation to aid the analysis of safety-critical systems
>>Controlled Natural Languages

Andrei Carniel (andrei.carniel@ga.ita.br)
Prof. Dr. Celso Massaki Hirata (hirata@ita.br)

Schedule

- Controlled Natural Language proposal

Controlled Natural Language (CNL) proposal

CNL proposal

- The proposal is based in Carvalho, G., Falcão, D., Barros, F., Sampaio, A., Mota, A., Motta, L., & Blackburn, M. (2014). There are some considerations added after reading the work Tommila, T., & Pakonen, A. (2014).

- The objective is to make a CNL proposal and not a formalization. The proposal will be based on published works that aim to formalize a CNL for requirements and safety-critical systems.

CNL proposal

Orientations:

- Write short sentences;
- Avoid using passive voice and high textual complexity;
- Avoid ambiguity words, such as: maybe, sometimes, consider, and other words that can induce to more than one interpretation;
- Each requirement need to have a **conditional clause** and an **action clause**;
- Sentences separated by COMMA;
- After using the verb SHALL is necessary to have one "verbAction" and one "verbComplement", if have more than one is necessary to add COLON after SHALL keyword and use the COMA to delimit the element;
  - For example: *When the system mode is idle, and the coin sensor changes to true, the coffee machine system shall: reset the request timer, assign choice to the system mode.*
- Use DOT to finish the requirement.

(Carvalho, G., Falcão, D., Barros, F., Sampaio, A., Mota, A., Motta, L., & Blackburn, M., 2014)

CNL proposal

Carvalho, G., Falcão, D., Barros, F., Sampaio, A., Mota, A., Motta, L., & Blackburn, M. (2014), does not specify an explicit statement for keywords but use the "when" as the same purpose as Tommila.

Keywords (Tommila, T., & Pakonen, A., 2014):

- **When:** event or action, e.g. "When the operator pushes a button, …".
- **While:** represents an interval of time, e.g. "While the plant is in the shut-down state, …", or "While the field operator is working inside the containment, …..".
- **Whenever:** same as "when" with the idea that there may be many occurrences or the event?
- **Where:** a conditional set of entities irrespective of time, e.g. "Where a motor-operated valve is installed inside the containment, it shall …." (see Mavin et al. 2009, 2010).
- **If:** a) Statement of a logical implication (causality) irrespective of time, e.g. "If the pressure is higher than x, the vessel may burst". b) Decision making in the course of an action, e.g. "While performing a periodic check, if the filter is dirty, the technician shall replace it".
- **Then:** a) Consequence of logical implication, e.g. "If ….., then ….". b) The temporal order of events and states, e.g. "….and then …".

CNL proposal

Verbs to use:

- Is;
- To add;
- To assign;
- To be (mandatory);
- To become;
- To change;
- To reset;
- To subtract;
- Are;
- Shall (mandatory);

(Carvalho, G., Falcão, D., Barros, F., Sampaio, A., Mota, A., Motta, L., & Blackburn, M., 2014)

CNL proposal

Logical operators:

- And (AND)
- Or (OR)
- Not (NOT)
- Shall (SHALL) - to specify a verb action and a verb complement
- ":" (COLON) - used after "shall" to presents a set of actions
- "," (COMA) - to separate the elements
- "." (DOT) - to finish the requirement

(Carvalho, G., Falcão, D., Barros, F., Sampaio, A., Mota, A., Motta, L., & Blackburn, M., 2014)

CNL proposal

Comparisons to use (relational operators):

- Less than/Lower than
- Less than or equal to
- Equals
- Greater than or equals to
- Greater than/More than/Higher than;
- Different

(Carvalho, G., Falcão, D., Barros, F., Sampaio, A., Mota, A., Motta, L., & Blackburn, M., 2014)

According to Oxford Dictionary (2020) we must use:

- Less for uncountable things and few for countable things.
- More for a larger number or amount, instead of greater.

CNL proposal

Operators suggested by Tommila, T., & Pakonen, A. (2014):

- Logical: not, and, or, xor, implies, iff
- Relational: >, <, >=, >=
- Arithmetic: +, -,*, /
- The lowest level refer to variables and constants. For example: ((LEVEL_T10 <= 85) & MAN_RESET).

CNL proposal

Rules (Carvalho, G., Falcão, D., Barros, F., Sampaio, A., Mota, A., Motta, L., & Blackburn, M., 2014):

- **Rule 1.** The verbs to add, to assign, to reset, and to subtract shall be used only in action statements.
- **Rule 2.** The verbs to be, to become, and to change shall be used only in the conditions of a requirement.
- **Rule 3.** Concerning the verbs to add, to assign, and to subtract, all thematic roles associated with action statements are mandatory.
- **Rule 8.** Concerning the verb to add, the structure is: to add X to Y.
- **Rule 9.** Concerning the verb to assign, the structure is: to assign X to Y.
- **Rule 10.** Concerning the verb to be, the structure is: to be X.
- **Rule 11.** Concerning the verb to become, the structure is: to become X.
- **Rule 12.** Concerning the verb to change, the structure is: to change from X to Y or to change to Y.
- **Rule 13.** Concerning the verb to reset, the structure is: to reset X.
- **Rule 14.** Concerning the verb to subtract, the structure is: to subtract X from Y.

CNL proposal

NOT USED rules (Carvalho, G., Falcão, D., Barros, F., Sampaio, A., Mota, A., Motta, L., & Blackburn, M., 2014):

- ***Rule 4***. Concerning the verb to reset, the thematic role TOV is not used, and the other thematic roles associated with action statements are mandatory. We assume to reset to be equivalent to assigning a standard value. This standard value may be dependent on the context (e.g., in the context of boolean values, reset is equivalent to assign false).
- ***Rule 5***. Concerning the verbs to be and to become, the thematic role CFV is not used, the thematic role CMD is optional, and the other thematic roles associated with conditions are mandatory. We assume to become X to be equivalent to changes from not X to X.
- ***Rule 6***. Concerning the verb to change, the thematic roles CFV and CMD are optional, and the other thematic roles associated with conditions are mandatory. The use of to change without the CFV is equivalent to the use of to become. For instance, changes to Xis equivalent to write becomes X. We also have one rule to avoid an incoherent use of the CMD thematic role.
- ***Rule 7***. Concerning the thematic role CMD, incoherent modifiers shall not be used. Thus, it is not possible to have greater than and lower than simultaneously for the same Condition Modifier (CMD) role as it does not make sense. Finally, the last rules concern the prepositions and number of complements that shall be used with each verb.

CNL proposal

Suggested structures (Tommila, T., & Pakonen, A., 2014):

Clause type	Examples of sentence structures
Variable value	<variable> is <value | range> <variable> is <comparison> <value> <variable> is <comparison> <variable>
Operational state	<entity> is in <operational state> state
Operating mode	<entity> is in <operating mode> mode
Variable value change	<variable> becomes <comparison> <value> <variable> becomes <comparison> <variable>
Operational state change	<entity> enters <operational state> state
Operating mode change	<entity> enters <operating mode> mode

CNL proposal

Suggested structures (Tommila, T., & Pakonen, A., 2014):

Clause type	Examples of sentence structures
Action	<entity> performs <action> <entity> starts <activity> <entity> <verb> <entity> <entity> sets <property> of <entity> <entity> sets <variable> to a value that is <comparison> <value> <entity> sets <entity> to <operational state> state
Maintaining	<entity> keeps <property> of <entity> in the range <range>
Performing	<entity> performs <activity>
Within	<entity> performs <action> within <duration>

CNL proposal

Suggested structures (Tommila, T., & Pakonen, A., 2014):

Clause type	Examples of sentence structures
Within	<entity> performs <action> within <duration>
Period	<entity> performs <activity> for a period of <duration>
Before	<entity> enters <operational state> state before <event> <entity> enters <operational state> state within <duration>
After	<entity> performs <action> after <event | duration>
Never	<entity> never performs <action>

CNL considerations

Tommila, T., & Pakonen, A. (2014) cite a Carvalho work published in 2013. Maybe that is the reason for the similarity in requirements constructions. According to Google Scholar Tommila has 14 citations.

Example of CNL requirements of Tommila, T., & Pakonen, A. (2014):

Informal requirement	CNL expression
Liveness	
When the tank level exceeds 85 cm, the flush command is given after a 10 s delay.	While condition (MAN_RESET) is false, condition (LEVEL_T10 > 85) is true leads eventually to condition (FLUSH).
The flush command is memorised.	Condition (SR_001.Q1) is true after condition (LEVEL_T10 > 85) is true until condition ((LEVEL_T10 <= 85) & MAN_RESET) is true.
After the tank level returns to allowed limit, the operator is able to manually reset the flush command.	While condition (LEVEL_T10 > 85) is false, condition (MAN_RESET) leads to condition (NOT FLUSH).
Safety	
The flush command is never given without the tank level having exceeded the allowed limit.	Condition (FLUSH) is false before condition (LEVEL_T10 > 85).

CNL considerations

Restrictions for verbs that Carvalho suggest, such as:

- ***Rule 1***. The verbs to add, to assign, to reset, and to subtract shall be used only in action statements.
- ***Rule 2***. The verbs to be, to become, and to change shall be used only in the conditions of a requirement.

- There is no delimited structure in Carvalho articles, but it is possible to identify the elements of requirement structure. Carvalho fragments the requirement in two main parts: condition and action.

CONDITION	ACTION
<keyword/grammar> <variable/entity> <verb/comparison> <value>,	<variable/entity> <verb> <value/action> <variable/entity>

Examples of CNL requirements

LINK FOR THE TRAIN DOOR EXAMPLE: HTTPS://DOCS.GOOGLE.COM/SPREADSHEETS/D/1CE5TPS8HHCIG6LZK0BBK44JXH-FUEPEOUZRULFT_S1Q/EDIT?USP=SHARING

CNL considerations

REQ001 – When the system mode is idle, and the coin sensor changes to true, the coffee machine system shall: reset the request timer, assign choice to the system mode. (Carvalho, G., Falcão, D., Barros, F., Sampaio, A., Mota, A., Motta, L., & Blackburn, M., 2014)

Condition				Action			
Keyword	**Variable**	**Verb/ Comparison**	**Value**	**Variable/ Entity**	**Verb**	**Value/ Action**	**Variable/ Entity**
When	the system mode	is	idle				
, and	the coin sensor	changes to	true				
				, the coffee machine system	shall:		
					reset		the request timer
					, assign	choice	the system mode.

CNL considerations

REQ002 – When the water pressure becomes higher than or equal to 900, and the pressure mode is low, the Safety Injection System shall assign permitted to the pressure mode.

(Carvalho, G., Falcão, D., Barros, F., Sampaio, A., Mota, A., Motta, L., & Blackburn, M., 2014)

Condition				Action			
Keyword/ Grammar	**Variable**	**Verb/ Comparison**	**Value**	**Variable/ Entity**	**Verb**	**Value/ Action**	**Variable/ Entity**
When	the water pressure	becomes higher than or equal to	900				
, and	the pressure mode	is	low				
				, the Safety Injection System	shall assign	permitted	to the pressure mode.

CNL considerations

Trains door requirement: *When the door state is person in doorway, the train door controller shall not perform close door.*

Or

Trains door requirement: *When the door state is equals to person in doorway, the train door controller shall not perform close door.*

Condition				Action			
Keyword/ Grammar	**Variable**	**Verb/ Comparison**	**Value**	**Variable/ Entity**	**Verb**	**Value/ Action**	**Variable/ Entity**
When is	the door state	is	person in doorway	, the train door controller	shall not perform	close door.	
When is	the door state	is equals to	person in doorway	, the train door controller	shall not perform	close door.	

CNL considerations

Trains door requirement: "When the door position is fully open, or door position is partially open, the train motion shall not change from stopped."

Condition				Action			
Keyword/ Grammar	**Variable**	**Verb/ Comparison**	**Value**	**Variable/ Entity**	**Verb**	**Value/ Action**	**Variable/ Entity**
When	the door position	is	fully open				
, or	door position	is	partially open				
				the train motion	shall not change from	stopped.	

References

- Carvalho, G., Falcão, D., Barros, F., Sampaio, A., Mota, A., Motta, L., & Blackburn, M. (2014). NAT2TESTSCR: Test case generation from natural language requirements based on SCR specifications. *Science of Computer Programming*, *95*, 275-297. DOI: http://dx.doi.org/10.1016/j.scico.2014.06.007.
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